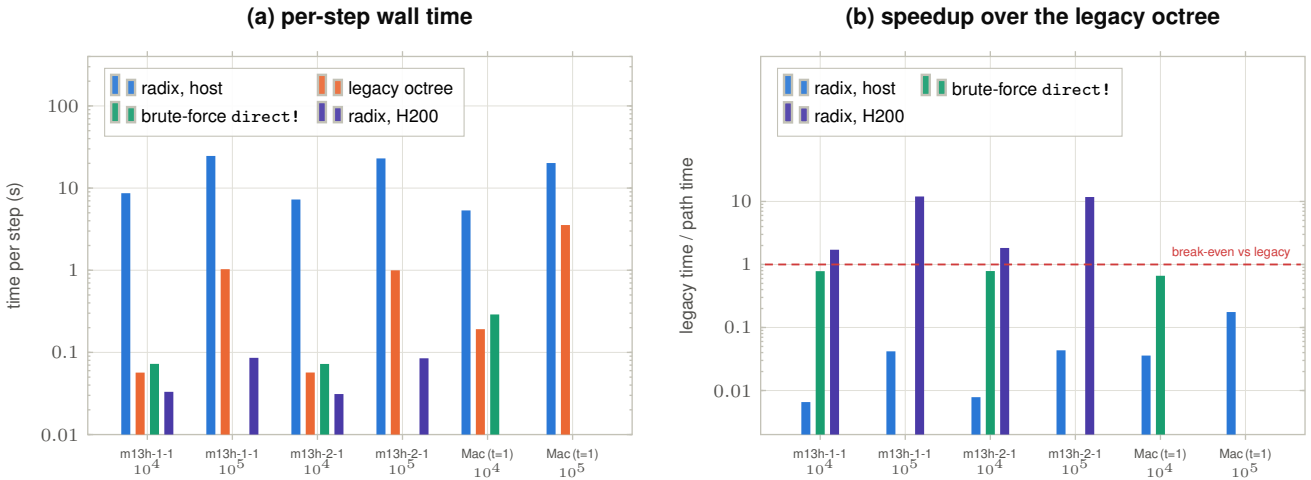
Fig. 8 — The end-user comparison: the radix path against the shipping legacy octree `fmm!`



Machines / regime: `m13h-1-1` and `m13h-2-1` (Julia 1.11.7, `threads=8`, NVIDIA H200) and a local Apple-silicon Mac (Julia 1.12.5, `threads=1`, no GPU); $P = 4$, $\ell = 4$, $n \in \{10^4, 10^5\}$. Thread counts come from the run sidecar `.md` files — the CSVs omit them. **Data:** task 023, `production_integration/benchmark_023_{m13h-1-1,m13h-2-1,mecsrs-MacBook-Pro-188.local}_*.csv` (tables `fig08a_*`, `fig08b_*`). **Reads as:** the *device-resident* radix path is a real end-user win at $P = 4$ — $1.71\times$ the legacy octree at $n = 10^4$ growing to $11.99\times$ at $n = 10^5$ on `m13h-1-1` ($1.83\times$ / $11.76\times$ on `m13h-2-1`). The *host* radix path is not: it is $152\times$ slower than legacy at $n = 10^4$ and $24\times$ slower at $n = 10^5$, and at $n = 10^4$ it is $\approx 120\times$ slower than brute-force `direct!` — i.e. slower than doing no FMM at all. That is consistent with the `019b` decision: the legacy octree remains the CPU default and the radix cache exists to feed the GPU loop. **There is no roadmap item to optimize the host radix path. Caveats — this is not an apples-to-apples comparison, in four independent ways, none of them disclosed in task 023.** (1) *Threading.* `src/fmm.jl` has nine `Threads.@threads`/`@spawn` sites; the radix host path (`src/translate_batched{,_resident}.jl`) has none. Legacy is multithreaded and host radix is not, which is why the gap collapses from $152\times$ at 8 threads to $28\times$ at 1 thread on the Mac. BLAS threads were never set or recorded. (2) *Different admissibility criteria.* Radix uses `ConstantPAnalyticStencil` at `stencil_epsilon=1e-4` on a fixed uniform 2^4 grid; legacy uses `multipole_acceptance=0.4` with `leaf_size=20` on a shrunk adaptive octree. Same P, structurally different work. (3) *No error check at the benchmarked n.* The 023 script computes no error norms and never uses the `direct!` result as a reference; the only accuracy evidence in task 023 is at $n = 500$. (4) *Asymmetric timing boundary, against legacy.* The legacy timing includes both octree builds and the interaction-list build on every call (`src/fmm.jl:1037-1058`), while the radix *step* excludes its one-shot cache construction, recorded separately in Fig. 3(c) — so the boundary favours radix and legacy still wins by $152\times$; the bias understates how slow the host path is. **Protocol:** times are the *minimum* over repetitions (radix 5 jittered steps, legacy 3 reps); `direct!` is a *single* un-repeated `@elapsed` and was run only at $n = 10^4$. Missing bars are unrun configurations, not zeros. **Scope:** every bar is a complete evaluation — far field *and* nearfield — on both paths.